

Integration

Rule: Increase the power by 1 and divide by the new power.

$y =$	$\int y dx =$
x^2	$\frac{x^3}{3} + c$
$12x$	$\frac{12}{2}x^2 + c = 6x^2 + c$
9	$9x + c$
$x^3 - 6x^2 + 8x - 5$	$\frac{x^4}{4} - \frac{6}{3}x^3 + \frac{8}{2}x^2 - 5x + c =$ $\frac{x^4}{4} - 2x^3 + 4x^2 - 5x + c$

Integration is the reverse of differentiation.

Example:

Find y if $\frac{dy}{dx} = x^2 - 6x + 2$ and $y = 4$ when $x = 3$.

$$y = \frac{x^3}{3} - \frac{6}{2}x^2 + 2x + c \quad \text{i.e.} \quad y = \frac{x^3}{3} - 3x^2 + 2x + c.$$

When $x = 3, y = 4$. So

$$4 = \frac{3^3}{3} - 3 \times 3^2 + 2 \times 3 + c$$

$$4 = 9 - 27 + 6 + c$$

$$4 = -12 + c$$

$$c = 16$$

So, $y = \frac{x^3}{3} - 3x^2 + 2x + 16$

Definite Integration

Example:

Find $\int_1^4 (2x-1)(x+2)dx$.

First expand the brackets: $\int_1^4 (2x-1)(x+2)dx = \int_1^4 (2x^2 + 3x - 2)dx$

Integrate the expression: $= \left[\frac{2}{3}x^3 + \frac{3}{2}x^2 - 2x \right]_1^4$

Substitute in the 2 limits and subtract:

$$= \left(\frac{2}{3}4^3 + \frac{3}{2}4^2 - 2 \times 4 \right) - \left(\frac{2}{3}1^3 + \frac{3}{2}1^2 - 2 \times 1 \right)$$

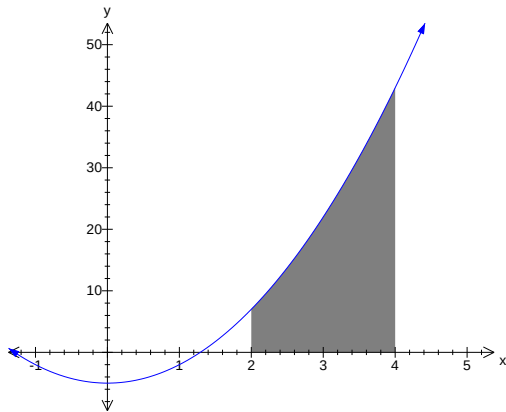
$$= 58\frac{2}{3} - \frac{1}{6}$$

$$= 58\frac{1}{2}$$

Finding areas

Integration can be used to find the area underneath a curve.

Example 1:

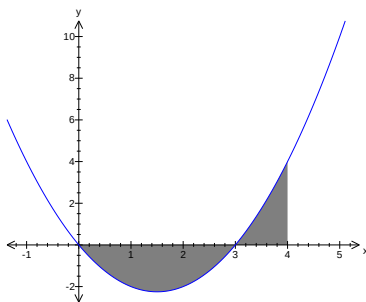


Find the area beneath the curve $y = 3x^2 - 5$ between the lines $x = 2$ and $x = 4$.

$$\begin{aligned}\text{Area} &= \int_2^4 (3x^2 - 5)dx = \left[x^3 - 5x \right]_2^4 \\ &= (4^3 - 5 \times 4) - (2^3 - 5 \times 2) \\ &= 44 - (-2) \\ &= 46\end{aligned}$$

Areas beneath the x-axis are negative. You need to calculate areas above and below the axes separately.

Example 2:



The diagram shows the curve $y = x(x - 3)$. Find the shaded area.

The curve crosses the x-axis at $x = 3$ (and at $x = 0$).

Area above the x-axis is

$$\begin{aligned}\int_3^4 x(x-3)dx &= \int_3^4 (x^2 - 3x)dx = \left[\frac{1}{3}x^3 - \frac{3}{2}x^2 \right]_3^4 \\ &= \left(\frac{1}{3}4^3 - \frac{3}{2}4^2 \right) - \left(\frac{1}{3}3^3 - \frac{3}{2}3^2 \right) \\ &= -2\frac{2}{3} - (-4.5) \\ &= 1\frac{5}{6}\end{aligned}$$

Area below the x-axis is

$$\begin{aligned}\int_0^3 x(x-3)dx &= \int_0^3 (x^2 - 3x)dx = \left[\frac{1}{3}x^3 - \frac{3}{2}x^2 \right]_0^3 \\ &= \left(\frac{1}{3}3^3 - \frac{3}{2}3^2 \right) - \left(\frac{1}{3}0^3 - \frac{3}{2}0^2 \right) \\ &= (-4.5) - 0 \\ &= -4.5\end{aligned}$$

So total area is $1\frac{5}{6} + 4\frac{1}{2} = 6\frac{1}{3}$.

Note:

To find the area between 2 curves use the formula:

$$\text{Area} = \int (\text{top curve} - \text{bottom curve}) dx$$

Questions:

1. (i) Integrate $\int (x+1)(x-2)dx$.
(ii) Evaluate $\int_{-1}^3 (x+1)dx$.
2. Calculate the area completely enclosed by the curve $y = x^3 - 3x^2 + 4$, the x -axis and the lines $x = 1$, $x = 4$.
3. Find the area of the finite region bounded by the curve $y = 12 - x - x^2$
4. Carry out the following integrations:
(a) $\int (x+3)dx$ (b) $\int 3x dx$ (c) $\int x^3 dx$.
and the x -axis.
5. Evaluate $\int_{-1}^0 (x^2 - 1)^2 dx$.
6. Calculate $\int_1^2 (x^2 - 3x)dx$. On a sketch graph shade an area corresponding to this integral.
7. Find the coordinates of the points where the line $y = x + 11$ meets the curve $y = (x+2)(x-5)$ and sketch the two graphs on one diagram.
Evaluate $\int_{-3}^7 [x+11 - (x+2)(x-5)] dx$.
On your diagram, shade the area represented by this integral.
8. $v = \int (3x^2 + 4x - 2)dx$.
(i) If $v = 3$ when $x = 0$, find v as a function of x .
(ii) Hence calculate the value of v when $x = 1$.
9. a) Find the coordinates of the points of intersection of the line $y = 2x$ with the curve $y = x^2 - 3x$.
b) Sketch the line and the curve for $0 \leq x \leq 6$.
c) Calculate the area of the region enclosed between the line and the curve.